Earth-State Embeddings:

a self-supervised codec of the North Atlantic, its transport read-out, and what a transformer forecasts from it

blauewelt/earth — technical report

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Abstract

We compress the observed state of each North Atlantic grid cell — ocean reanalysis, Argo climatology, wind stress, sea-surface temperature — into a 32-dimensional embedding with a self-supervised per-pixel codec, and train a transformer to predict the embedding forward at five-day cadence. **Codec.** Read out by an attention head over the individual pixels of the 26.5°N section, the frozen embeddings predict RAPID overturning transport at $r = 0.67$ (two codec seeds, year-blocked k-fold); the same head fed raw section values reaches 0.66, so the representation adds nothing measurable for the current-state read-out. Wind stress accounts for most monthly skill (a wind-only ridge reaches 0.53); the 16 Argo levels move skill to the density-driven MOVE transport at 16°N; neither codec capacity (0.92M to 40.7M) nor training length (50k to 1M steps) moves the probe. **Forecaster.** Under a training pool that excludes every window touching a held-out year, stage-2 heads from 7.6M to 400M parameters reach their best held-out one-step loss inside the first ∼2,000 of 200,000 steps and worsen thereafter, the best level lying in 0.58–0.62 of the persistence error across a 53× span in capacity. Rolled twelve months from held-out starts, the 206.7M head's field anomaly correlation decays from 0.61 at five days to 0.41 at ten, ≈ 0.1 from two months on and −0.03 at a year; it beats raw persistence at every lead but the last and damped persistence at none. Its squared-error score is negative (mean MSSS −0.44 against climatology) because it emits anomalies at 78% amplitude at 10% correlation — an identity reproduces all 73 leads to 0.0135, and the same states rescaled would score ≈ +0.02. Skill that survives past a few pentads lives in sea-surface temperature; 32 of 40 channels score nothing at any lead. Transport at 26.5°N shows no measurable skill at roughly nine effective starts. Every rolled number is a single seed.

1 Data

The working tensor covers 0–70°N, 100°W–20°E on a 0.25° grid (281 × 481 cells, 86,698 ocean) at five-day cadence from 1982-01 to 2024-12 (3,142 bins), with 40 channels per cell: surface current speed, log mixed-layer depth and sea-surface height from the GLORYS reanalysis; temperature and salinity at 16 pressure levels from the Roemmich–Gilson Argo climatology (native 1°, monthly, from 2004); NCEP Reanalysis-1 wind stress and its within-bin variability (four channels); OISST sea-surface temperature. Channels absent before their record begins enter as missing tokens. Earlier codec generations used 1° monthly tensors with 12–25 channels, on the same window or globally; they are identified by channel count below. The transport truth is the RAPID-MOCHA overturning at 26.5°N (2004–), deseasonalised; MOVE at 16°N is a second target. Everything cited here — tensors, weights, result bundles — is public (Appendix A).

2 The codec and its read-outs

Model. A masked autoencoder over one pixel's channels at one time bin produces a d_z-dimensional embedding; a 3 × 3-patch variant sees the pixel's neighbours. The working codec on the 0.25° pentad tensor is 512-wide, 12 layers, decoder width 256, d_z=32, patch 1, 37.976M parameters, trained 200k steps at batch 512 with 2009, 2017 and 2023 held out of training and of the anomaly statistics. Embeddings are probed for transport with a year-blocked k-fold: a ridge on the section-mean embedding (*pooled*), or an attention head over the section's individual pixels (*unpooled*). Field reconstruction is scored as the percentage improvement over persistence, one bin ahead.

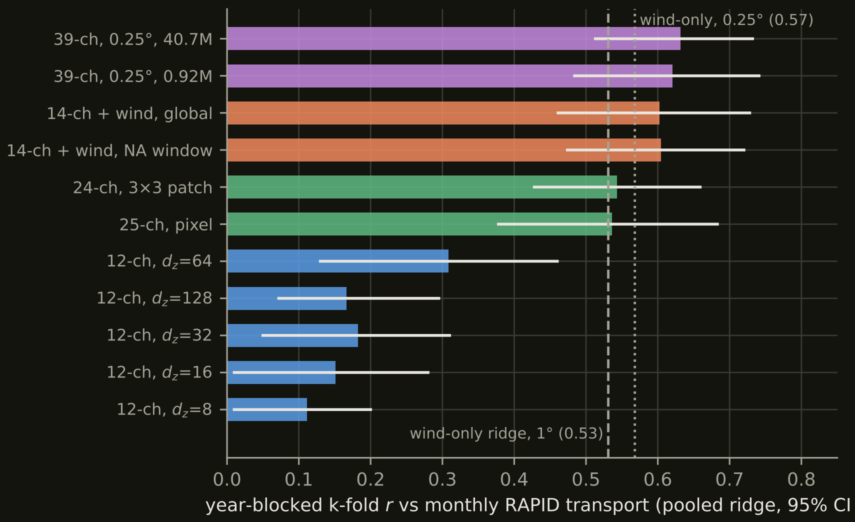

Figure 1: Pooled k-fold correlation with monthly deseasonalised RAPID transport by codec generation. Blue: the 12-channel width sweep; orange: with wind stress; green: 24–25 channels with 8 Argo levels; violet: the 0.25° tensor with 16 levels and SSH at two capacities.

What moves the transport read-out. Figure 1 is the pooled ranking across generations. Adding wind stress to a 12-channel tensor takes the pooled probe from 0.31 to 0.60; a wind-only ridge with no embedding reaches 0.531 on the 1° tensors and 0.568 on the 0.25° one, so most monthly skill at 26.5°N is Ekman physics. Training on the global ocean instead of the window costs the Atlantic read-out nothing (0.602 against 0.604). Bottleneck width matters little for reconstruction (saturating by d_z=16–32) and peaks at d_z=64 for the probe, turning over at 128 against ∼220 truth months per fold. On the 0.25° tensor a 0.92M pilot reads 0.620 and the 40.7M codec 0.631 — 44× the parameters buys +0.011, inside seed noise (±0.05 on the pooled probe) — and a codec trained from 50k to one million steps is flat from the first probe to the last. Neither capacity nor training length is what stands between this codec and a better transport number.

Pooling was hiding the signal, and the codec is not what recovers it. The same frozen embeddings read three ways (Figure 2): an MLP on the section mean does no better than the ridge, so the ridge was never the limit as a function class; the attention head over the unpooled section beats every pooled probe and lifts the 3 × 3-patch codec to 0.690 [0.570, 0.781] (RMSE 2.02 Sv against a 2.79 Sv target). Geostrophic transport is the east–west contrast across the section, and a section mean is the statistic in which that contrast cancels. The same head fed raw section anomalies instead of embeddings completes the attribution:

tokens carry	raw data	codec embedding
single pixel	0.613 [0.48, 0.73]	0.617 [0.49, 0.73]
3 × 3 neighbourhood	0.659 [0.55, 0.75]	0.690 / 0.654 (two codec seeds)

Unpooling the section is the whole of the improvement (pooled 0.54 → head 0.65+); the 3 × 3 receptive field adds ≈ +0.045 and does so on raw data alone; pretraining adds +0.013 at two seeds, which is nothing against a head seed spread of 0.036. On the 0.25° monthly tensor the margin is +0.034 (0.662 against 0.628, one seed). At pentad cadence (n=1,459 transport bins) three unpooled read-outs — the frozen monthly codec (0.691), a codec trained on the pentad tensor itself (0.680), and raw fields with no codec (0.683) — land within 0.011 of one another, while the pooled ridge (0.660) sits below the pentad wind bar (0.670). For the current-state transport read-out, read-out design and not representation learning is what moved the number.

Where the deep channels go. Extending Argo from 4 to 16 levels did not raise the RAPID probe but roughly doubled skill at MOVE — a deep, density-driven transport — from 0.111 to 0.206 [0.044, 0.346],

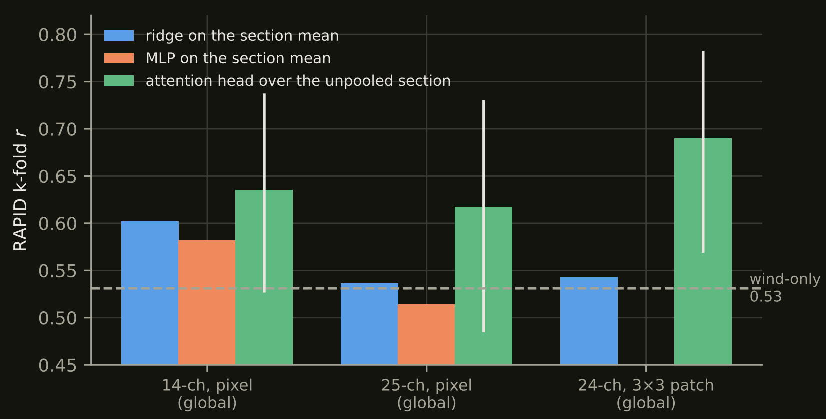

Figure 2: Same frozen embeddings, same folds, three read-outs. Head bars carry 95% intervals.

the first multi-target result whose interval excludes zero, with the 18-month low-passed correlation rising from 0.38 to 0.62. At 16°N the wind-only ridge is negative (−0.376). One task-blind representation serves a wind-dominated and a density-dominated target when the read-out respects the structure of the question.

Event anatomy. Out of fold, the winter 2009–10 collapse — the sharpest event in the RAPID record — is captured at 16% of its amplitude without wind channels, 50% with them, 59% by the 25-channel codec and 51% by the 40.7M 0.25° codec. Average correlation and event capture are different capabilities; the 25-channel codec has the best dip and the second worst monthly r.

Quantizing the embedding. A finite-scalar-quantized bottleneck (FSQ: each of d_z=6 dimensions rounded to 8, 8, 8, 5, 5 or 5 levels, one 2^{16}-way code per pixel and bin, a LayerNorm bound on the pre-quantization activation) trained from a random initialisation collapses on the pentad tensor: the encoder becomes input-independent within a few thousand steps (fitted level-spread 0.638 → 0.005 by step 7,500, twice), and an unbounded 8-level lattice on d_z=32 drifts to pre-quantization magnitudes $\sim 10^4$ against a lattice radius of 2, wearing its eight levels as a one-bit sign code. Switching the same lattice on at step 200k of a finished continuous d_z=6 codec and training 60k further steps keeps the activations input-dependent throughout (level-spread 0.97 → 0.80), at an 18% reconstruction tax (0.270 against 0.229) and a per-channel one-step error of 0.742 against 0.681 (persistence 0.843). The quantized codec reads RAPID at 0.588 [0.534, 0.641] against the continuous 0.579 [0.525, 0.627] through the unpooled head, one seed each — indistinguishable, and both below the raw 3×3 control at this width (0.693).

3 The forecaster

Model and protocol. A causal transformer over a window of K=144 consecutive embeddings at one pixel (two years) plus a 145-point spatial stencil on a spiral ring predicts the embedding one bin ahead, with input noise at 0.7 of the persistence scale, gradient clipping at 128, batch 256, learning rate 10^{-3} decaying exponentially (half-life 40k steps, warm-up 2k). The loss is dense over all 144 window positions, so the pool admits only windows that touch no held-out bin anywhere the forward pass can see (209,549,066 windows: 2,417 end-times × 86,698 pixels); it certifies this by brute force before training. The held-out one-step loss is the z-space MSE relative to persistence on held-out windows (denominator 21.44621). Widths: 256×8 (7.598M), 512×12 (40.388M), 1024×16 (206.659M), 1280×20 (399.948M); all on the frozen 37.976M codec and its embedding of the tensor.

A *roll* advances every ocean pixel one bin at a time from a true 144-bin context, feeding predictions back, for 73 steps (365 days), decodes through the codec and scores standardised anomalies per pixel and channel on three scopes: gate (600 random pixels plus the section), corridor (the fastest quarter of the window by current speed, dilated two cells, plus the section; 30,158 pixels), window (all pixels). Starts are three per held-out year, nine in all. Per lead: $\mathrm{MSSS}_x = 1 - \mathrm{MSE}/\mathrm{MSE}_x$ against climatology (zero anomaly), raw persistence, and damped persistence (per-pixel AR(1) fitted on training years);

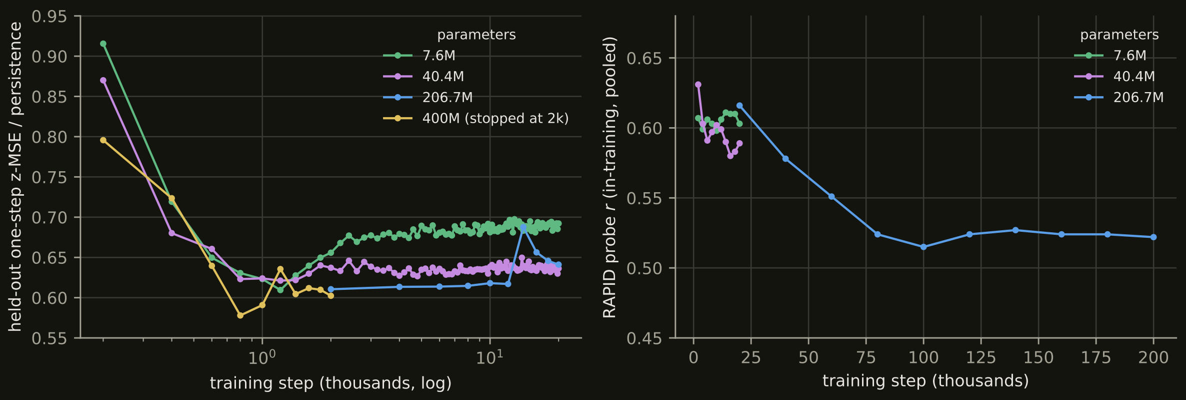

Figure 3: Left: held-out one-step ratio over the first 20,000 steps, by width. Right: the in-training RAPID probe over the full run.

params	best held-out ratio	at step	ratio at 20k	ratio at end	train ratio at end
7.598M	**0.6095**	1,200	0.692	0.692 (20k)	0.146
40.388M	0.6212	1,200	0.636	0.636 (20k)	0.086
206.659M	0.6105	$\leq$ 2,000	0.641	0.686 (200k)	0.023
399.948M	0.5779	800	—	0.602 (2k)	0.171

Table 1: Held-out one-step ratio by width. The 206.659M run is recorded every 2,000 steps, so its minimum is at or before its first record; the others, recorded every 200, bottom out at 1,200.

anomaly correlation ACC; amplitude ratio a. The transport read-out regresses the rolled section states onto RAPID and reports r in three lead bands.

The held-out minimum is early at every width. Figure 3 and Table 1. All four widths reach their best held-out ratio inside 2,000 steps and get worse afterwards; the best levels span 0.578–0.621 across 53$\times$ in parameters. At step 20,000 the 7.598M arm (0.692) is 0.051 worse than the 206.659M arm (0.641), and the ordering is not monotone in width (40.388M reads 0.636). Width buys how fast an arm reaches its minimum and how badly it then degrades, not the level of the minimum. The in-training transport probe separates the arms the other way: 7.598M holds 0.598–0.611 across ten probes, 40.388M drifts to 0.589, and 206.659M falls from 0.616 at 20k to 0.522 at 200k. The training loss keeps falling throughout (to a ratio of 0.023 at 200k for the 206.659M head, a 30$\times$ train/held-out gap): the model memorises the training years long after it has learned anything that transfers.

The twelve-month roll. The 206.659M head at its 200k end state, rolled from the nine held-out starts (Table 2, Figure 4). Corridor anomaly correlation is 0.606 at five days, 0.412 at ten, 0.265 at fifteen, 0.20 at a month, near 0.1 from two months on, -0.03 at a year. The head beats raw persistence at 72 of 73 leads (mean $\mathrm{MSSS}_{\mathrm{pers}}$ $+0.204$) and damped persistence at none: $\mathrm{MSSS}_{\mathrm{damped}}$ runs from -0.011 at five days to -0.719 at a year (mean -0.461; gate -0.416, window -0.420). A per-pixel AR(1) decay toward climatology, fitted on training years, is a better forecast of this field than the transformer at every horizon.

The negative score is amplitude, not sign. For a forecast with correlation ACC and amplitude ratio a against a zero-anomaly reference, $\mathrm{MSSS}_{\mathrm{clim}} = 1-(1+a^2-2a\,\mathrm{ACC})$ identically. From the head's own per-lead ACC and a this reproduces the 73 measured values to a mean absolute error of 0.0135 (Figure 5). The head emits anomalies at $a \approx 0.78$ while its correlation is 0.1; rescaled to $a = \mathrm{ACC}$ the same states would score ACC^2 per lead, a corridor mean of about $+0.02$. Whether a calibration fitted on training-year starts transfers to held-out starts is untested.

Per channel. Only eight of the forty channels are scoreable. Sea-surface temperature holds correlation longest (0.76 at five days, 0.34 at 30 and at 90) and is the only channel whose $\mathrm{MSSS}_{\mathrm{clim}}$ stays positive

lead (days)	$\mathrm{MSSS_{clim}}$	$\mathrm{MSSS_{pers}}$	$\mathrm{MSSS_{damped}}$	ACC	a
5	+0.365	+0.194	−0.011	0.606	0.692
10	+0.109	+0.252	−0.131	0.412	0.673
15	−0.172	+0.178	−0.354	0.265	0.761
30	−0.168	+0.252	−0.244	0.204	0.672
60	−0.320	+0.201	−0.342	0.107	0.690
90	−0.322	+0.211	−0.333	0.132	0.725
180	−0.354	+0.337	−0.357	0.153	0.778
270	−0.479	+0.227	−0.479	0.009	0.720
365	−0.719	−0.101	−0.719	−0.031	0.822
mean of 73 leads	−0.439	+0.204	−0.461	0.105	0.780

Table 2: The 206.659M head rolled from nine held-out starts, corridor scope. Gate and window scopes: mean $\mathrm{MSSS_{clim}}$ −0.395 and −0.401, mean ACC 0.131 and 0.121. Single seed.

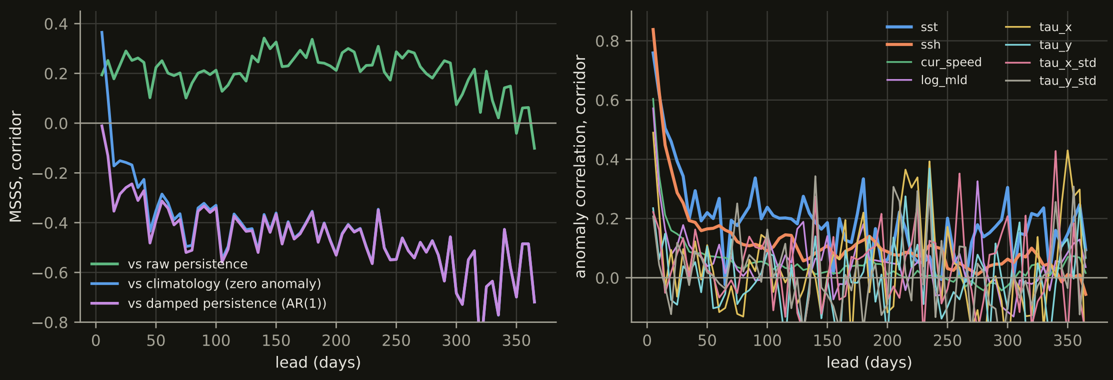

Figure 4: Left: the rolled head against its three references, corridor scope. Right: anomaly correlation against lead per scoreable channel; the 32 Argo channels have no scored samples at any lead.

beyond a few pentads; sea-surface height is sharpest at one step (0.84) and below 0.2 by a month; current speed and mixed-layer depth are one-step quantities; the four wind channels are noise past one step (Table 3).

Transport. The unpooled read-out of the rolled section states against RAPID gives $r = +0.107$ over 5–90 days (n=162), −0.242 over 95–180 (n=129), +0.163 over 185–365 (n=150). With about nine effective independent starts these are inside the noise of zero. A separate 36-month free run from a context ending 2004-12 — inside the training years, so not a skill measurement — correlates 0.636 with the true transport at pentad resolution and −0.13 on an 18-month low-pass at 61% amplitude: what the head tracks lives at high frequency, and the multi-year band is not measured by this roll.

Statistical power. The pool has 2,417 distinct end-times; consecutive windows share 143 of 144 frames, and every pixel is a correlated view of the same temporal patterns. With 206M parameters that is ∼85,000 parameters per pattern, and the training loss falling long after the held-out loss has turned is the expected consequence. For field-level scores the effective sample count is in the thousands and differences can be resolved; for the transport target (RAPID ∼20 years, anomalies decorrelating over months to years) it is of order ten, and a band correlation of ±0.2 cannot be. Every rolled number here is one seed; rolled skill at this cadence has not been replicated, and no two rolled numbers can yet be called different.

4 What the results support

For the current-state transport read-out the codec is at parity with the raw fields it compresses; its value, if any, is as a substrate that a transformer can attend over and roll forward, and on that use the transformer has learned what it learns inside two thousand steps at any width, forecasts sea-surface

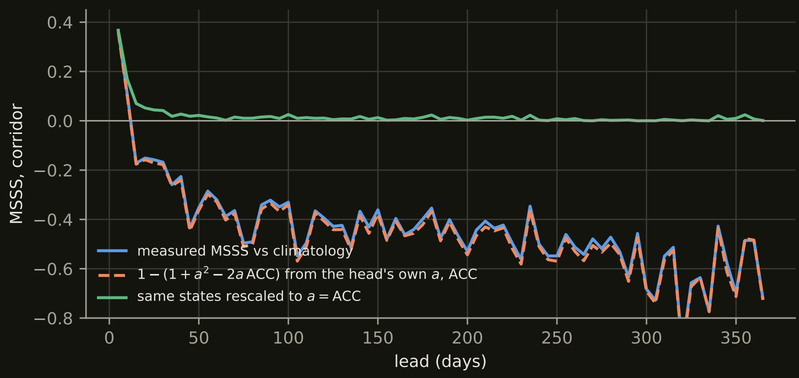

Figure 5: Measured corridor $\mathrm{MSSS}_{\mathrm{clim}}$ per lead, the value the identity predicts from the head's own ACC and a, and the value the same states would score at $a = \mathrm{ACC}$.

channel	ACC 5 d	ACC 30 d	ACC 90 d	mean $\mathrm{MSSS}_{\mathrm{clim}}$	mean $\mathrm{MSSS}_{\mathrm{damped}}$
`sst`	0.759	0.343	0.337	−0.165	−0.191
`ssh`	0.838	0.252	0.113	−0.499	−0.562
`cur_speed`	0.604	0.131	0.052	−0.591	−0.606
`log_mld`	0.572	0.178	−0.009	−0.464	−0.469
`tau_x`	0.490	0.097	0.022	−0.383	−0.386
`tau_y`	0.235	0.039	−0.017	−0.426	−0.428
`tau_x_std`	0.222	0.134	0.104	−0.409	−0.410
`tau_y_std`	0.207	0.070	0.057	−0.465	−0.466

Table 3: Per-channel corridor scores of the rolled head.

temperature and height for days to weeks, and does not beat a damped-persistence null at any lead. The next measurements are the ones that bound this from both sides: classical predictors (retrieval, a linear inverse model on the leading EOFs, in field space and in embedding space) through the identical battery, with block-bootstrap intervals; the early-stopped checkpoint and the 7.6M head rolled against the end state shown here; amplitude calibration fitted on training-year starts; and a codec trained on $\leq$ 2020 so that a terminal 2021–2024 test can be run once.

A Data and code availability

Code: `https://github.com/blauewelt/earth` (codec trainer `ml/train.py`, stage-2 trainer `ml/temporal.py` and its JAX port `ml/jaxport/`, evaluator `ml/rollout_spatial.py`). Result bundles are on the repository's `ml-metrics` branch (`probes-<n>.json`; the roll here is `probes-516.json`, the codec probe ladder is in the bundles named in `ml/EXPERIMENTS.md`). Weights are on the `model-checkpoints-v1` release (`run-415__pixelmae.pt`; `head-weights-e059-200k-window-s0.pt` and the `e060a`/`e060b` heads named likewise); the tensor on the `data-cache-v1` release and the Hugging Face dataset `chfrank/earth-tensors`; the codec's embedding of it on `embed-cache-v1`. Stage-2 figures and tables are generated by `ml/paper/make_figs.py` from `ml/paper/data/report_data.json`, built by `ml/paper/extract_data.py` from those bundles and the training records; codec probe numbers are transcribed from the archived per-run bundles, with the source named beside each in the figure script.

Sources: GLORYS12 and the GREP ensemble (E.U. Copernicus Marine Service Information, used under the CMEMS licence with credit to the service); the Roemmich–Gilson Argo climatology (Scripps; Argo data are collected and made freely available by the International Argo Program and the national programs that contribute to it); NCEP/NCAR Reanalysis-1 (NOAA PSL); OISST v2.1 (NOAA NCEI); the RAPID-MOCHA-WBTS transport time series (NERC, NSF, NOAA); the MOVE transport time series (NOAA).